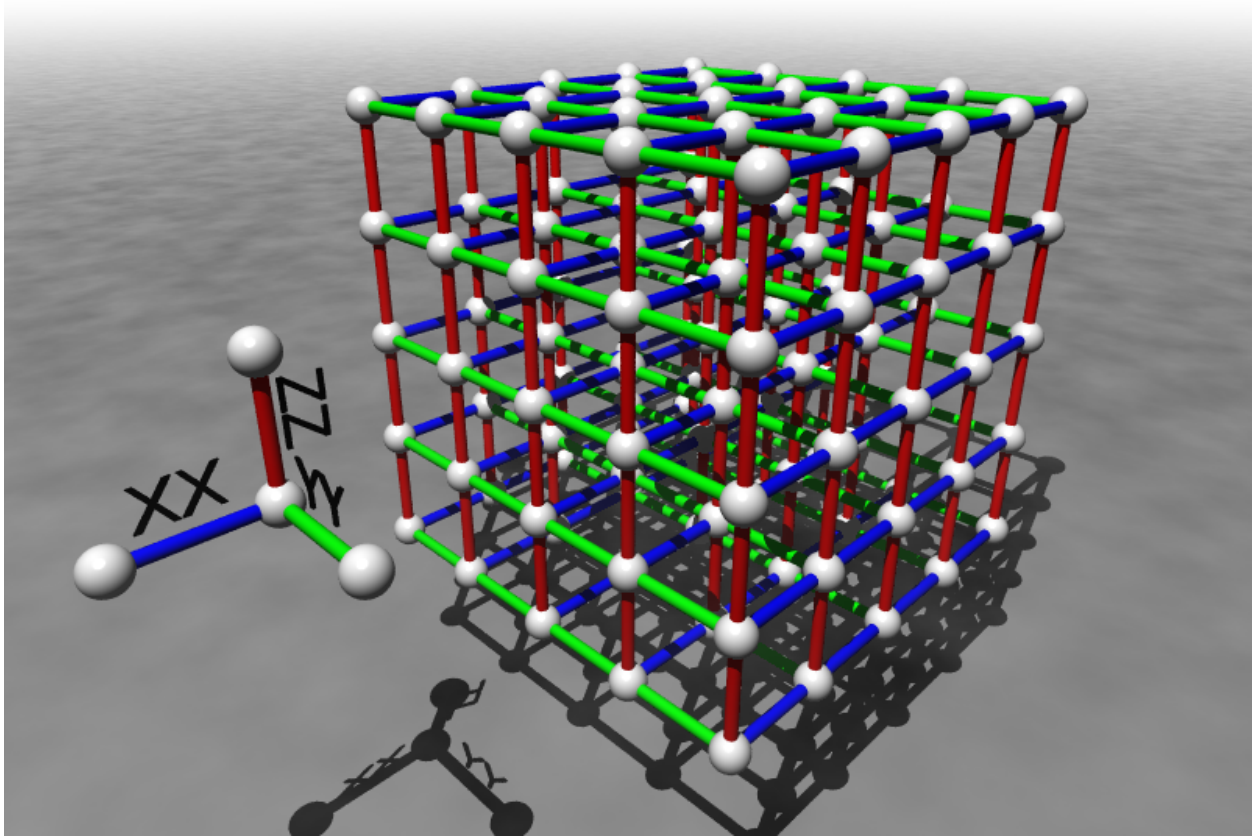
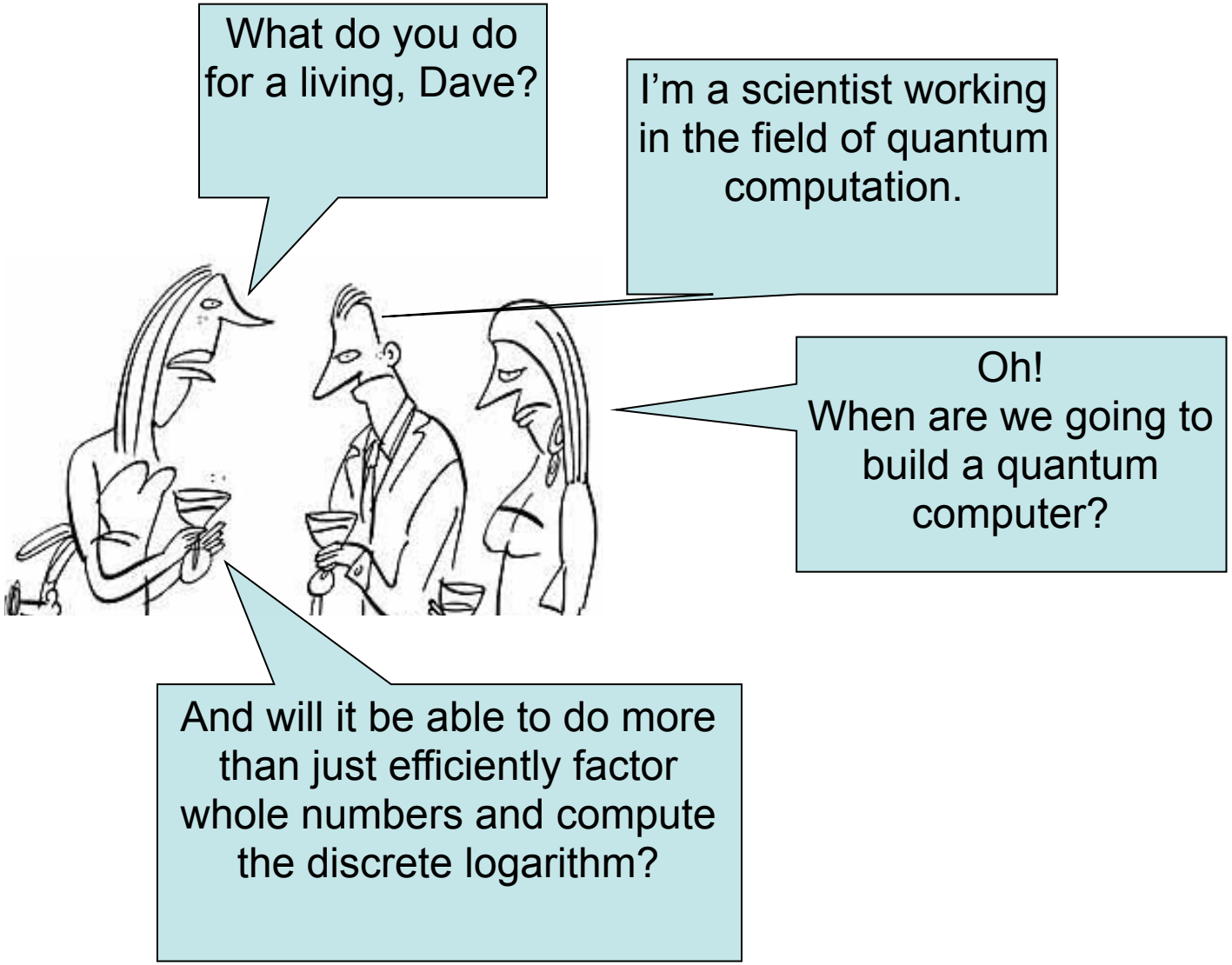
Towards Robust and Powerful Quantum Computers

Dave Bacon
University of Washington
Department of Computer Science & Engineering



The Question





Why the Confidence?

Why are we confident that large scale quantum computers can be built?

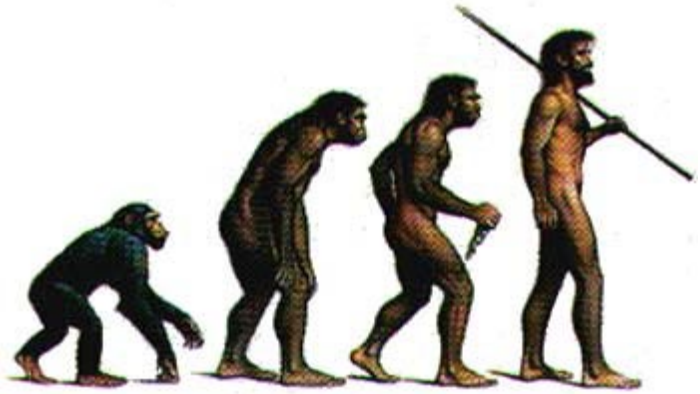


Why are we confident that quantum computers are worth studying and building?

A historical account: including the future!

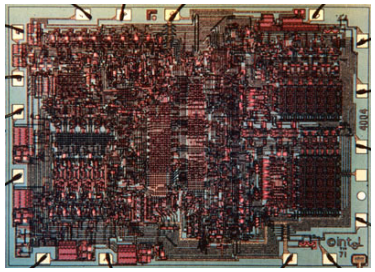


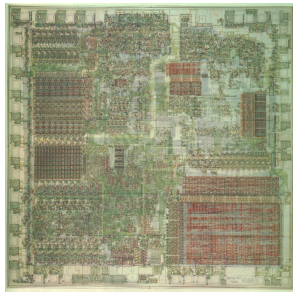
Where Are We Going?

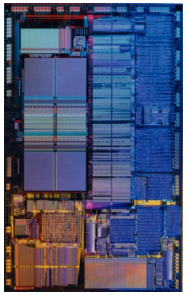




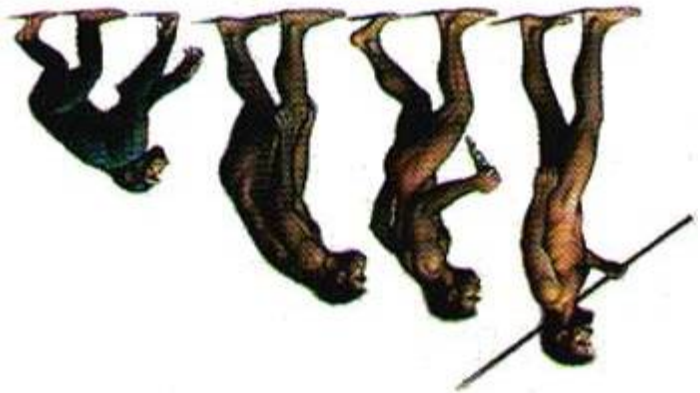




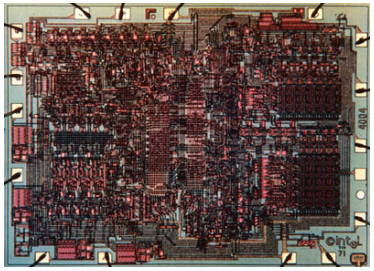


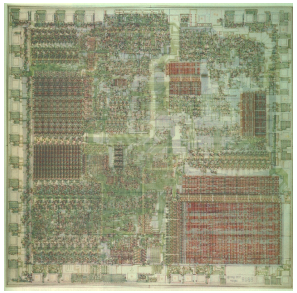


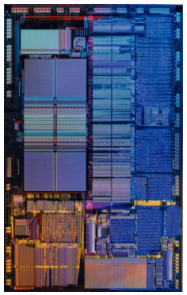
?



New Sizes, New Rules



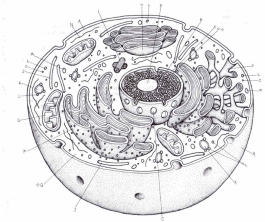




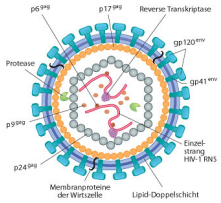
NOISY (FAULTY)

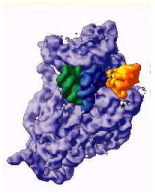
QUANTUM

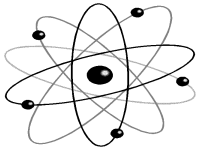
10,000nm 1,000nm 100nm 10nm 1nm 0.1nm









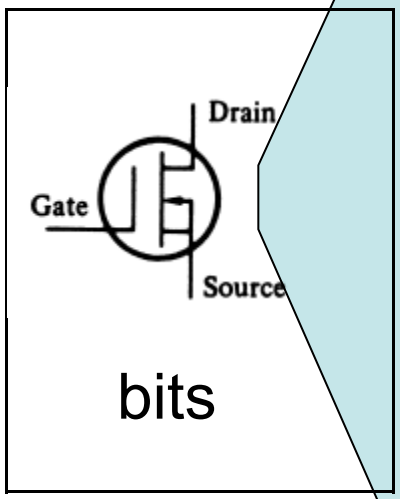


1. Can we build digital devices?
2. Quantum computers?

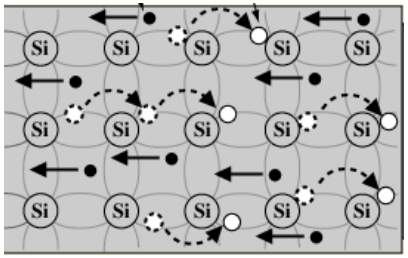
Bit, Pbit, and Qubit

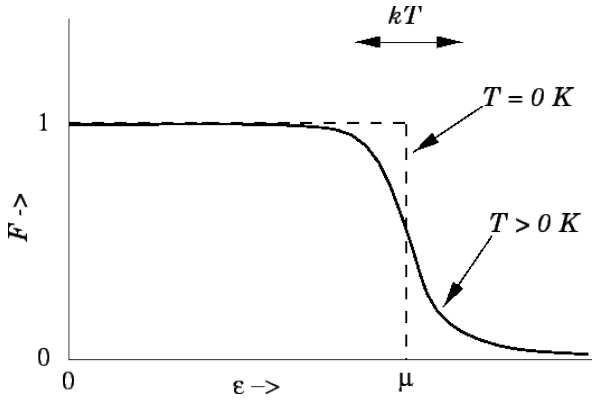
	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent 1 *1-p* percent	0 $\lvert\alpha\rvert^2$ercent 1 $\lvert\beta\rvert^2$ercent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$	$\begin{bmatrix} u & v \\ w & x \end{bmatrix}$
	deterministic	stochastic	unitary

There Are No Bits

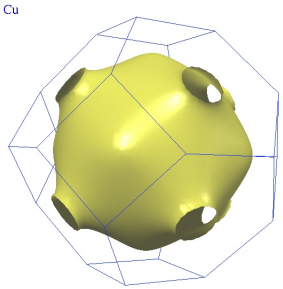


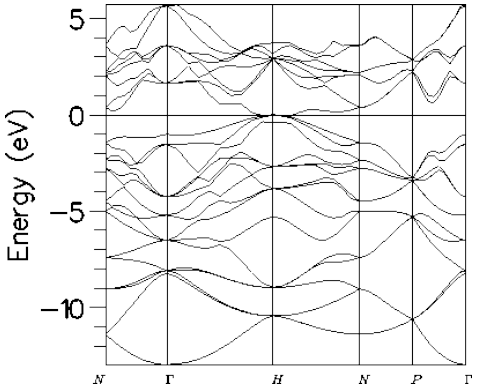
bits





pbits





qubits

Probabilistic Algorithms

Quantum Algorithms

Quantum Computers

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

3980750864240649373971255005503864911990643623425267084063851895754638895726176858331747277214610743530253622307197304822463291469530209711645985217113052071125636359039752 7

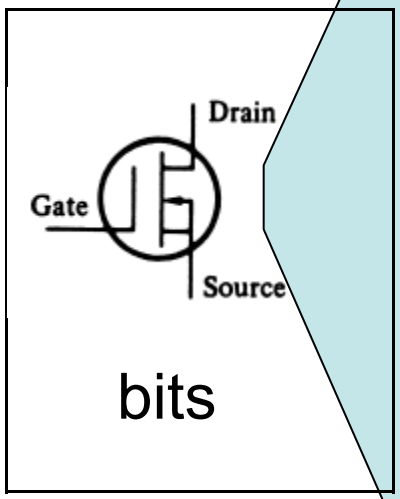


Peter Shor

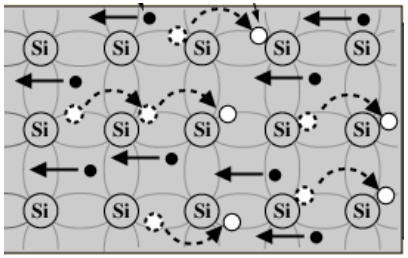
1994

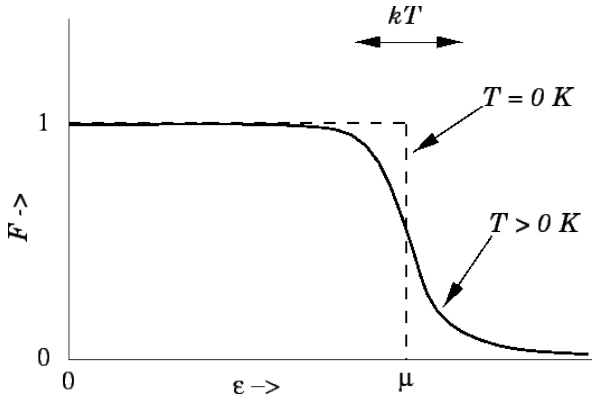
- Quantum computers can efficiently factor whole numbers (and compute discrete log.)
- No known classical efficient algorithm.
- Breaks RSA, Diffie-Hellman cryptosystems.

Qubits?

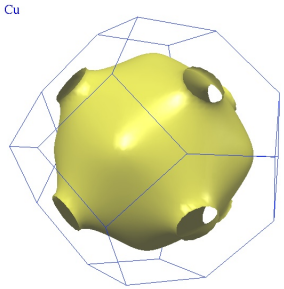


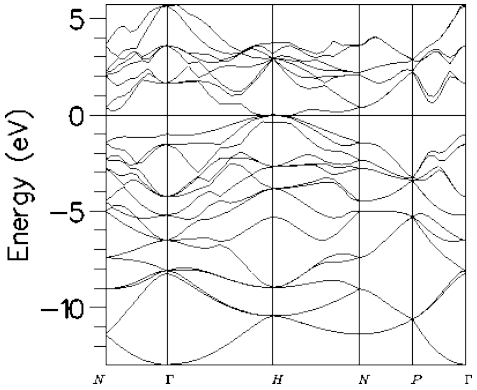
bits





pbits



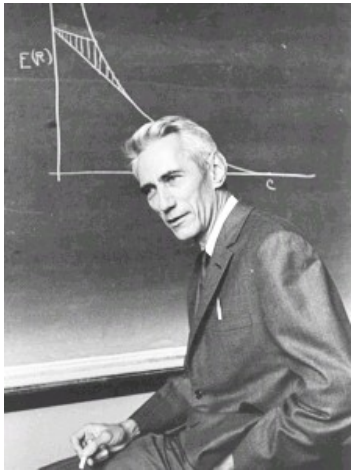


qubits

Quantum Algorithms

Can we exploit qubits in digital algorithms?

Noisy Pbits to Bits (Theory)



Claude Shannon

How to communicate in the presence of noise?

"A mathematical theory of communication" (1948)

Error correcting codes



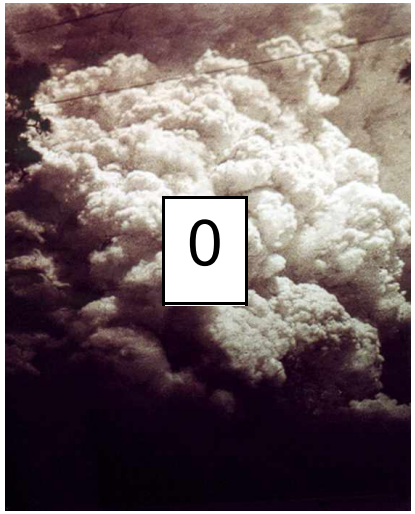
John von Neumann

How to compute when the subsystems of your computer can fail?

"Probabilistic logics and the synthesis of reliable organisms from unreliable components" (1956)

Fault-tolerant architectures

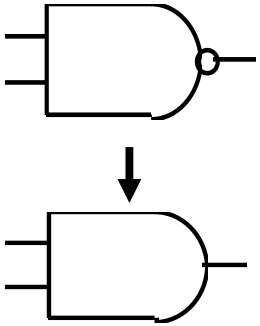
Fragile

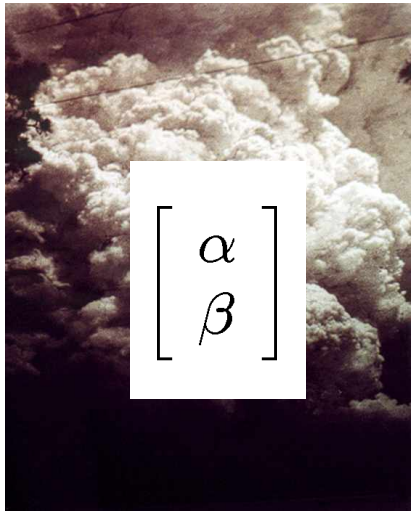


Classical Noise

$$\begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$$

Classical Faults



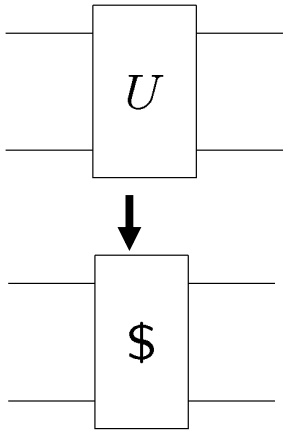


Quantum Noise

$$\begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix}$$

$$\begin{bmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{bmatrix}$$

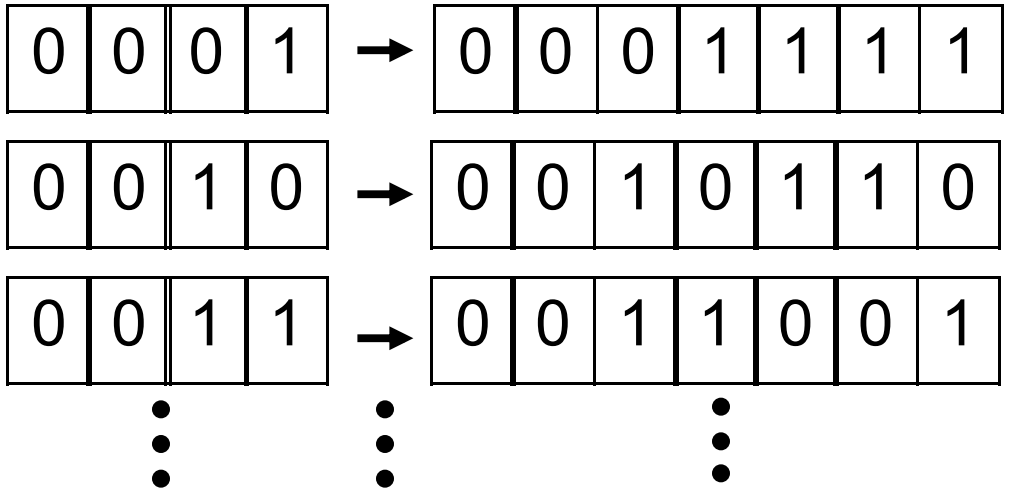
Quantum Faults



Error Correction

classical error correcting code:

(bits and pbits)





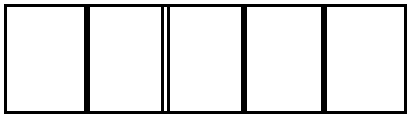
Peter Shor



Andrew Steane

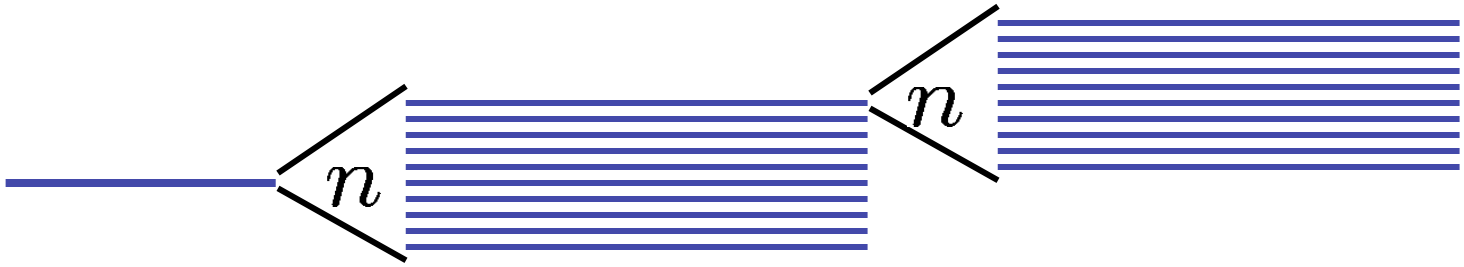
1995: quantum error correcting codes

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \rightarrow$$



encode quantum information across multiple quantum subsystems

Perfection Through Concatenation



failure probability

p *cp2* *c(cp2)2*

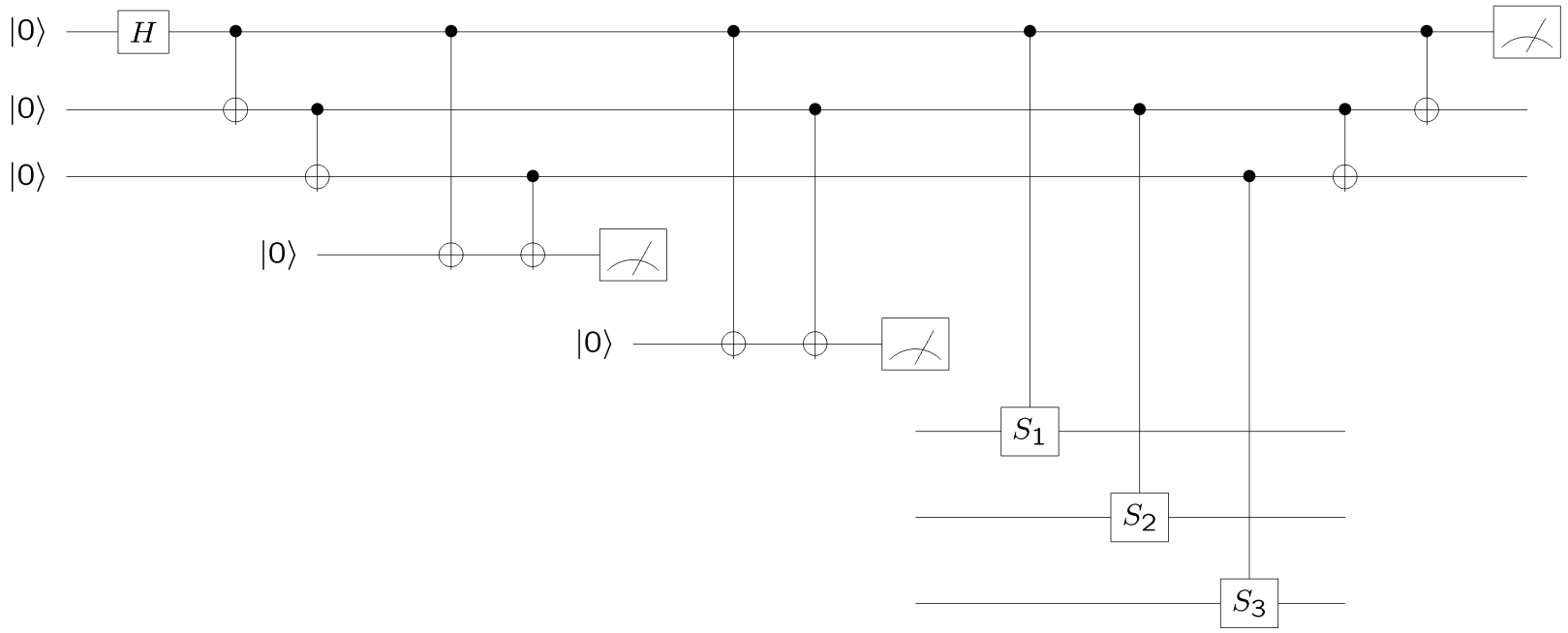
At kth level of concatenation, probability of an error which destroys the encoded quantum information is

$$p_{fail} = \frac{(cp)^{2^k}}{c} \qquad \frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k} \qquad p_{th} = \frac{1}{c}$$

and uses $N_{qubits} = n^k$ qubits

$$N_{qubits} = \left[\log_{\frac{p}{p_{th}}}\left(\frac{p_{th}}{p_{fail}}\right)\right]^{\log_2 n} = O\left(\text{poly}\log\frac{1}{p_{fail}}\right)$$

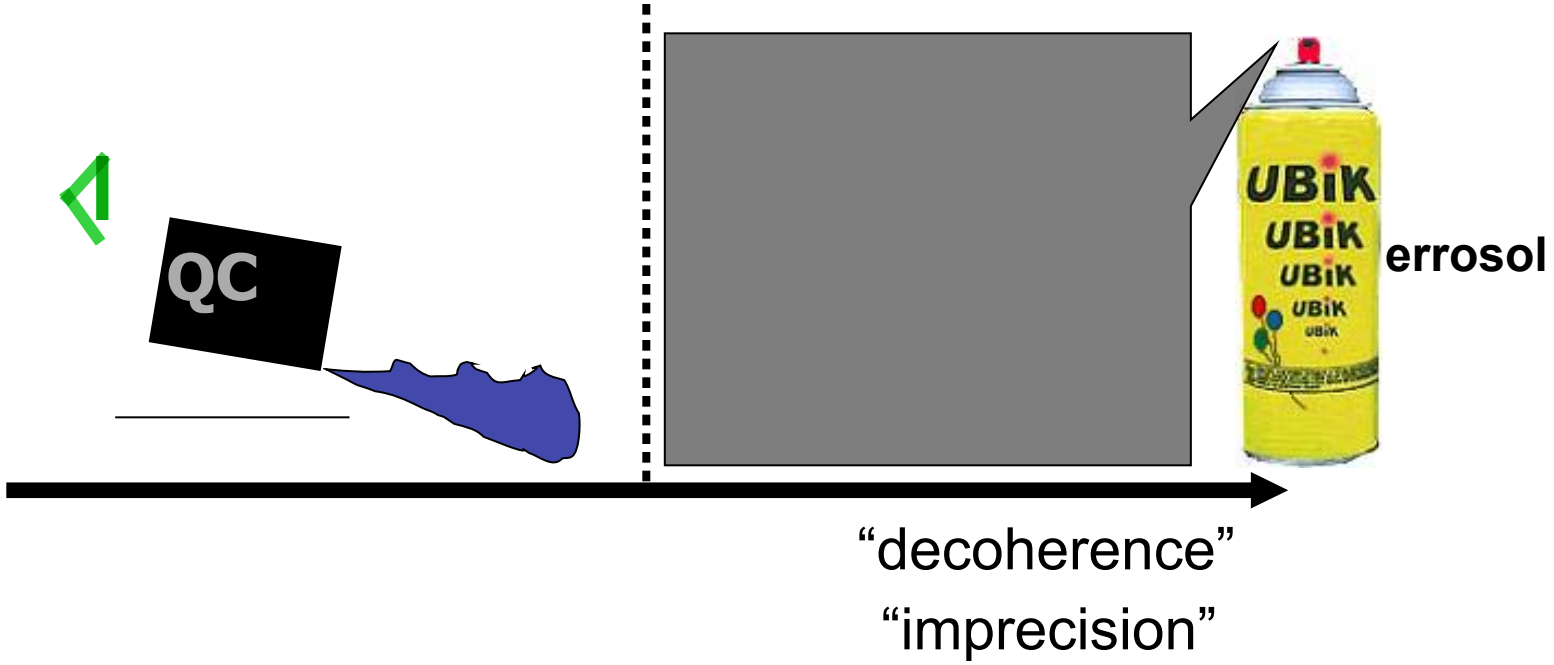
Fault-Tolerant Quantum Computation



Must design quantum circuits which succeed even when every element of the circuit (preparation, unitary gates, Measurements) can fail.

Quantum error correction + Fault-tolerant procedures

Threshold Theorem, 1996



Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Gottesman, Preskill]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

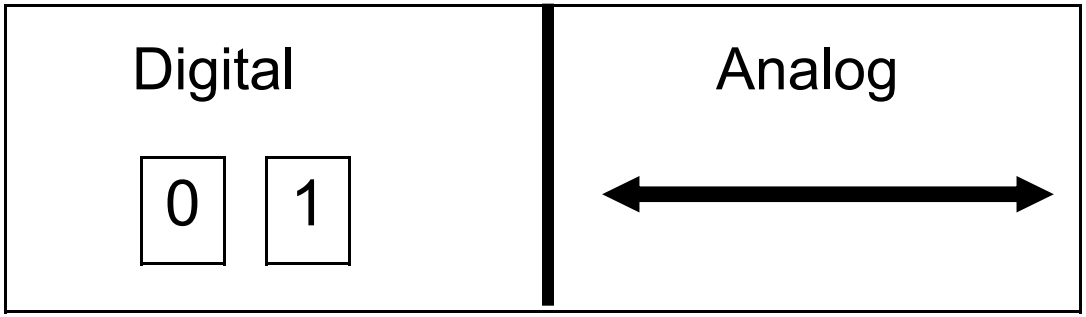
Reflection

No theory of fault-tolerant quantum computation

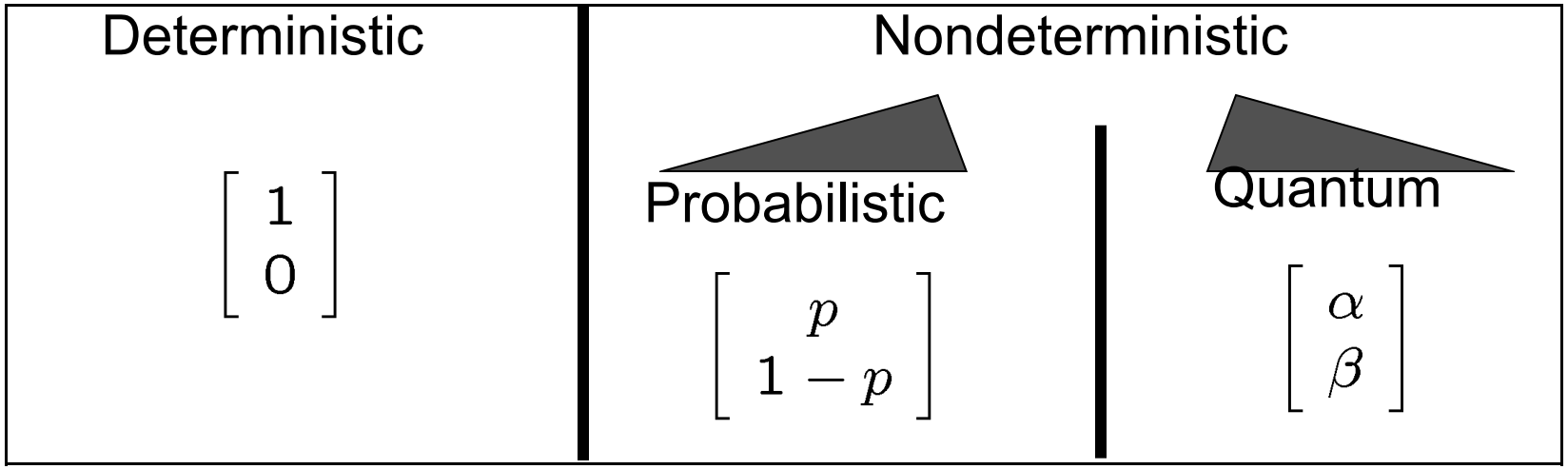
$\Rightarrow$

No such thing as a quantum computer

Common misconception



versus

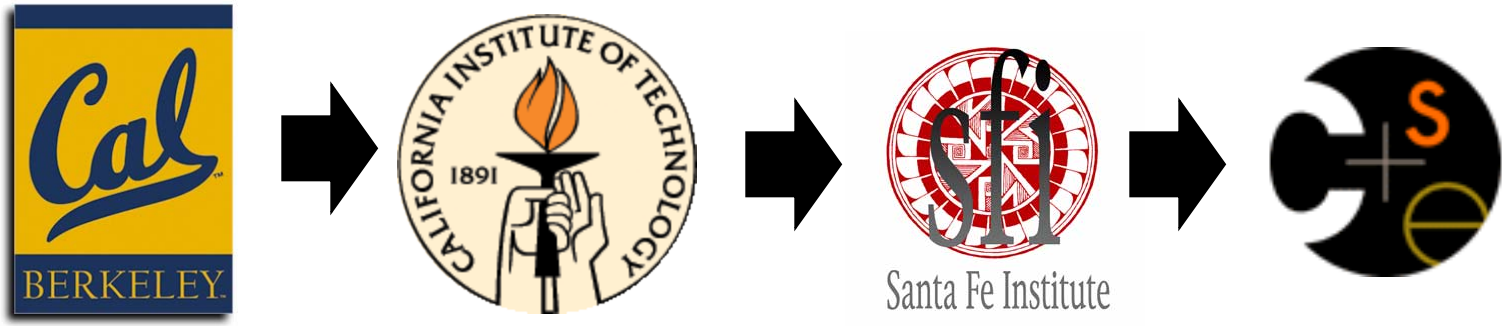


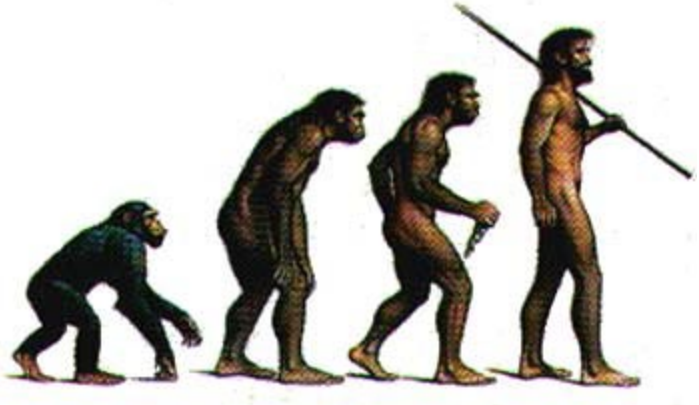
Reflection

Even with the theory of fault-tolerant quantum computation

$\Rightarrow$

Is quantum computation a viable technology?

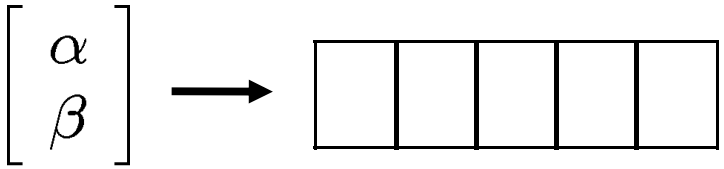


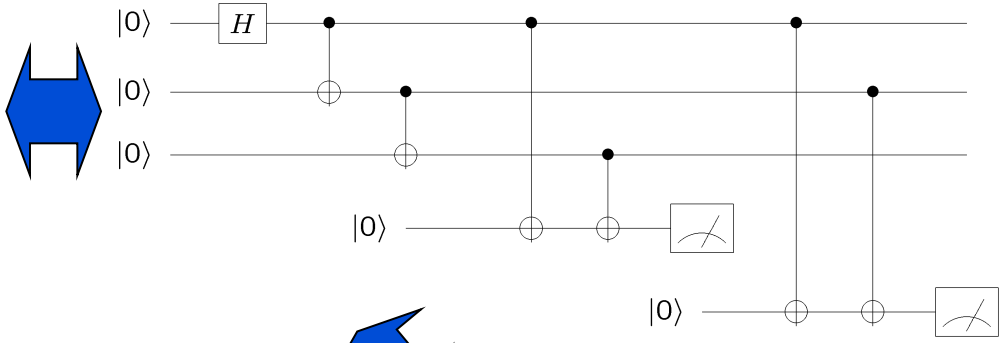


Peculiarities

quantum error correction

fault-tolerant methods





physical architecture

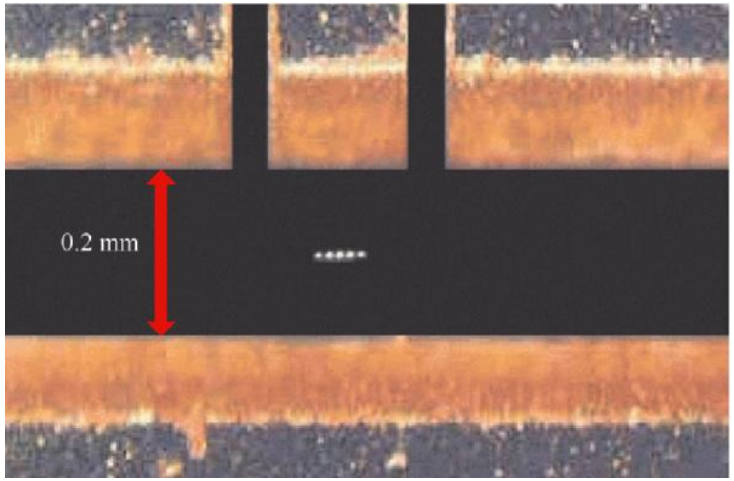
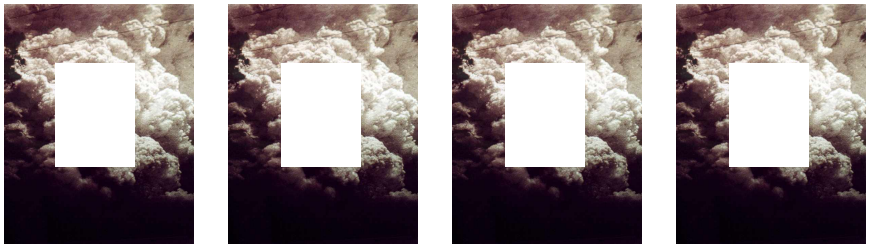
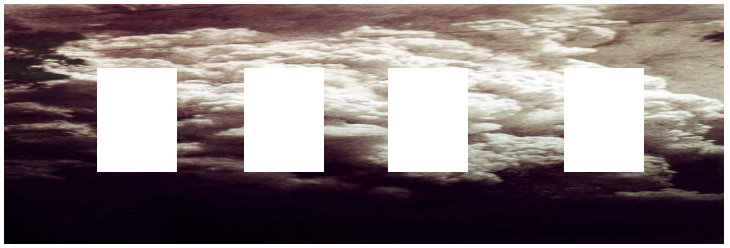


Photo from NIST Boulder

Decoherence-Free

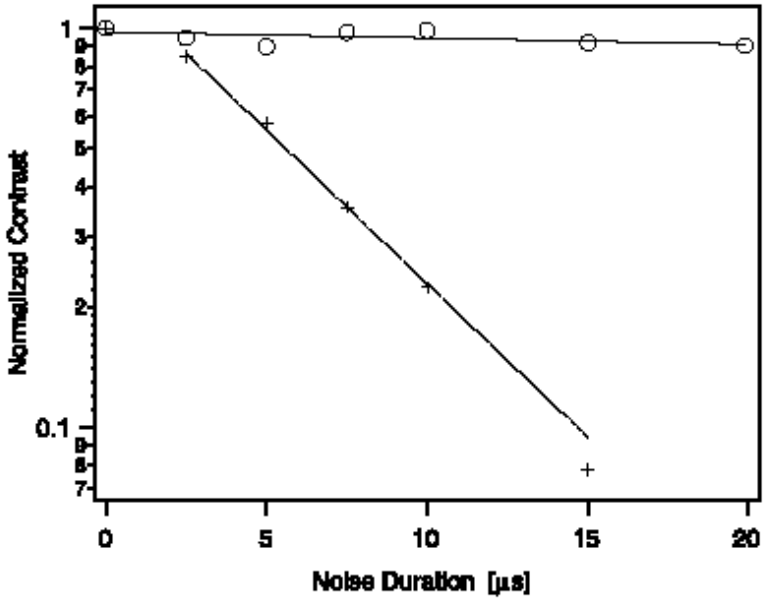


individual environments



collective environment

Use SYMMETRY of coupling to COMPLETELY eliminate effect of noise by encoding into a special quantum error correcting code (DFS)



REPORTS

A Decoherence-Free Quantum Memory Using Trapped Ions

D. Kielpinski,[1]* V. Meyer,[1] M. A. Rowe,[1] C. A. Sackett,[1]
W. M. Itano,[1] C. Monroe,[2] D. J. Wineland[1]

SCIENCE VOL 291 9 FEBRUARY 2001

Decoherence-Free QC

Symmetry breaking interactions enter at third order (robustness)

[Bacon, Lidar, and Whaley, *Phys. Rev. A* 60, 1944 (1999)]

Alternative symmetric decoherence mechanisms

[Lidar, Bacon, Kempe, and Whaley, *Phys. Rev. A* 63, 022306 (2001)]
[Lidar, Bacon, Kempe, and Whaley, *Phys. Rev. A* 63, 022307 (2001)]

Combining with quantum error correcting codes

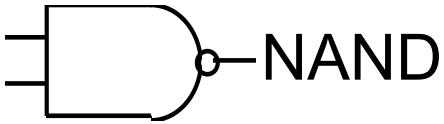
[Lidar, Bacon, and Whaley, *Phys. Rev. Lett.* 82, 4556 (1999)]

How to perform quantum computation on the information encoded in a decoherence-free subspaces?

Exchange Is Universal

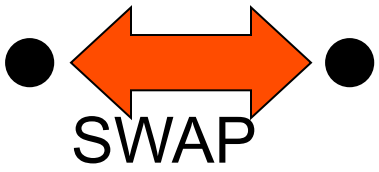
[Bacon, Lidar, and Whaley, *Phys. Rev. Lett.* 85, 1758 (2000)]
[DiVincenzo, Bacon, Kempe, Burkard, and Whaley, *Nature* 408, 339 (2001)]
[Kempe, Bacon, Lidar, and Whaley, *Phys. Rev. A* 63, 042307 (2001)]

Universal Classical Gate Set



(fan-out, fan-in)

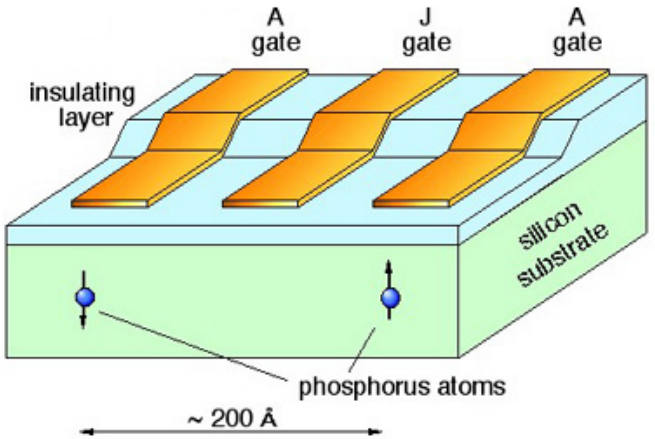
Universal Quantum Gate Set



exp(-i t0 SWAP)

exchange interaction alone is universal!

Just as NAND, NOR, use less resources, exchange interaction is least costly interaction in many solid state quantum comptuers



[Bacon, Kempe, DiVincenzo, Lidar, and Whaley, *Proc. Int. Conf. on Exp. Imp. of Quant. Comp.* Sydney, Australia (2001)]
[Kempe, Bacon, DiVincenzo, and Whaley, *Quant. Inf. & Comp.* 1, 33 (2001)]

But Still, a Technology?

Observation:



tum error correction is not that different from
ical error correction.

Observation:



Our most basic classical computational components
aturally robust (and often need only minor amounts
of classical error correction.)

Observation:



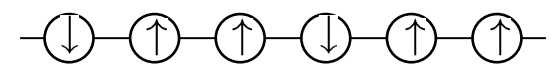
Device complexity of quantum error correcting schemes
ery high. "A quantum computer will spend most of it's
time performing error correction."

Question: What are the mechanism which allow robust classical computational components and can we port them to the quantum world?

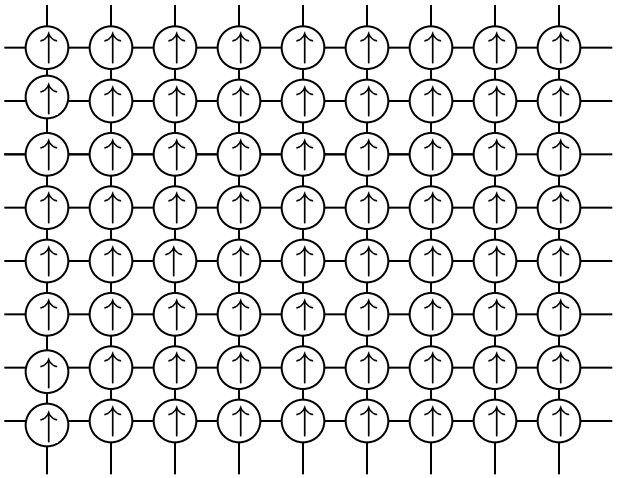


Three Storage Testbeds

Ferromagnetic Ising models:



one dimensional



two (and >) dimensional

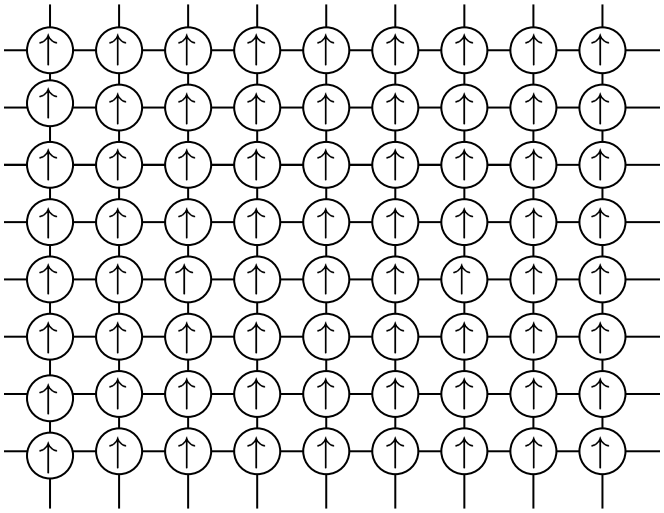
Toom's rule:

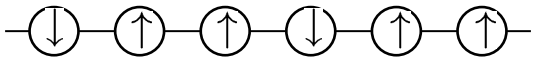
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Ferromagnetic Ising Models







Ising model

spins on edges of a lattice: $s_i \in \{+1, -1\}$

energy of a configuration:

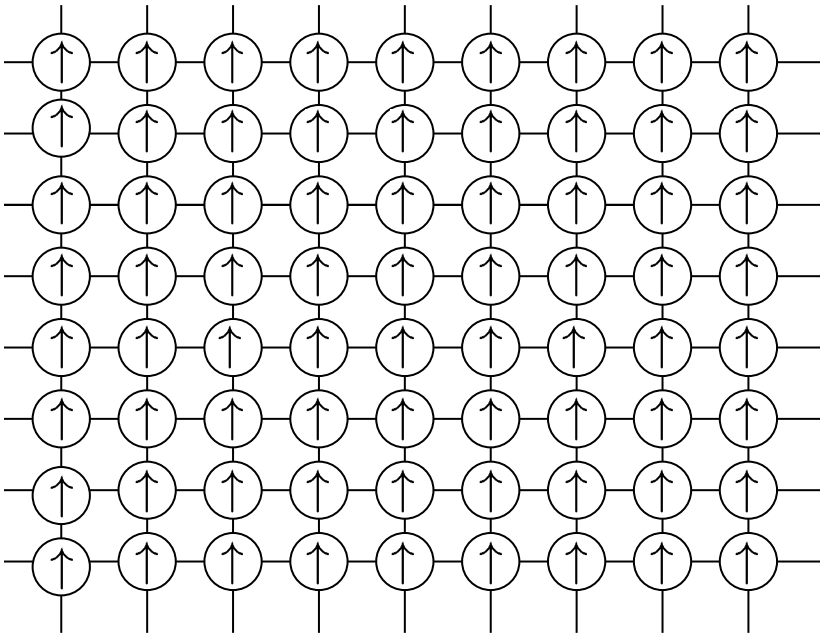
$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i \qquad J > 0$$

sum over neighbors

external magnetic field

$h = 0$ for now

Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

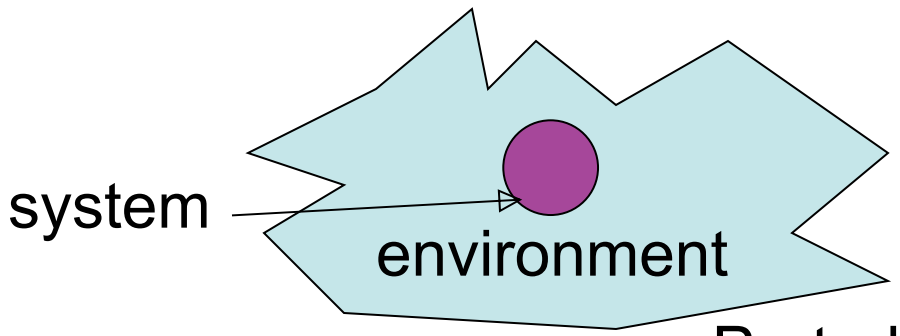
Update rule:

If flipping spin would decrease energy, flip spin

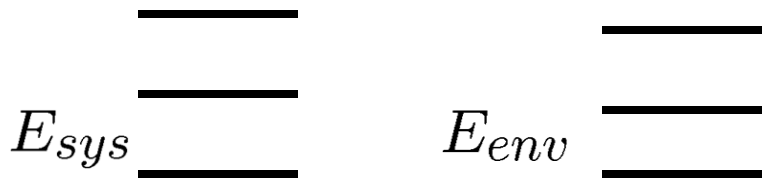
If flipping spin would increase energy, flip spin with probability

$\exp(-\beta \Delta E)$

Energetics of Decoherence

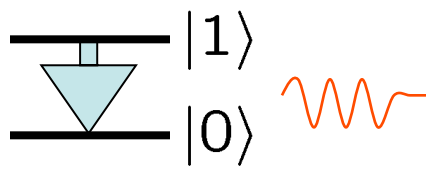


$$H = H_{sys} + H_{int} + H_{env}$$



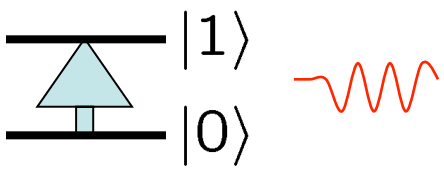
Perturbative regime: $E_{int} \ll E_{sys}, E_{env}$

Perturbative regime: decoherence conserves system and environment energy.



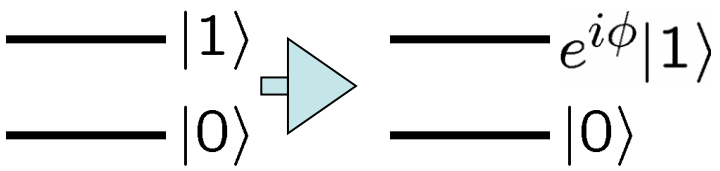
cooling

$\Delta E_{sys} < 0$



heating

$\Delta E_{sys} > 0$

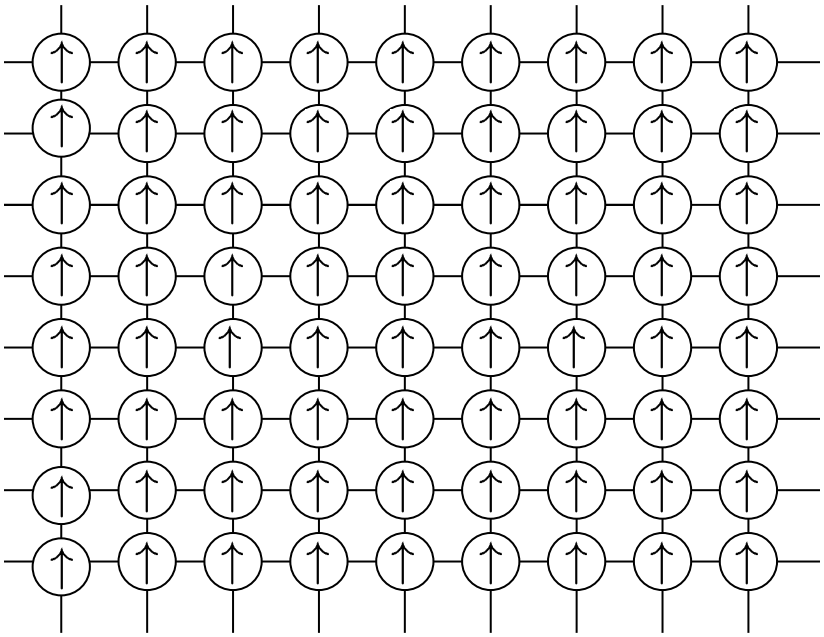


dephasing

$\Delta E_{sys} = 0$

by **cooling environment** we can eliminate heating

Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule (alternate with noise):

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

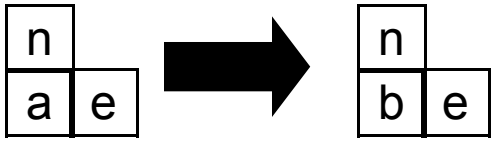
Toom's Rule



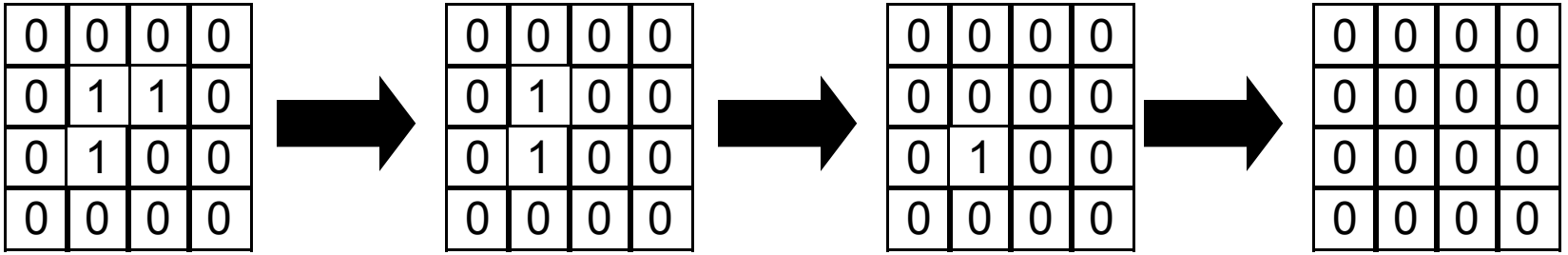
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Update rule:



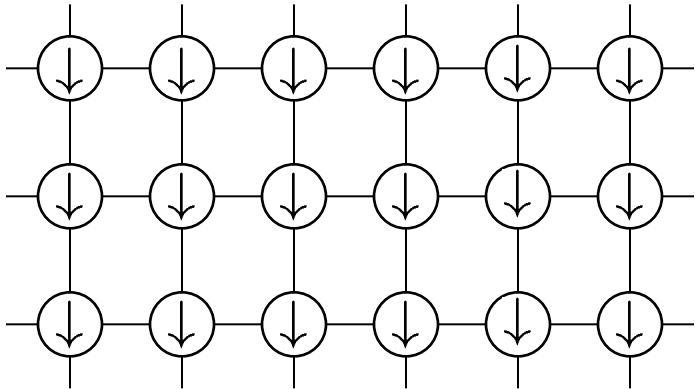
b=majority(n,e,b)



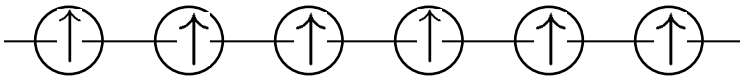
Update rule with noise: with some probability the update fails and a random state is assigned

Compare and Contrast

Encode information into the system and watch what happens



0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0



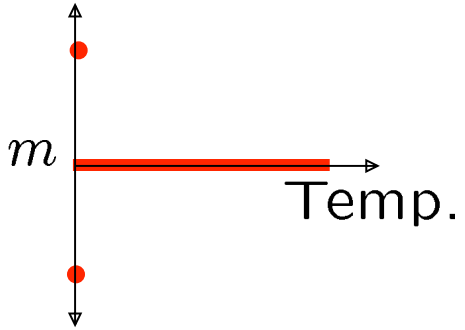
Redundancy code.

"Order parameter" is magnetization: $m = \frac{1}{N} \sum_i s_i$

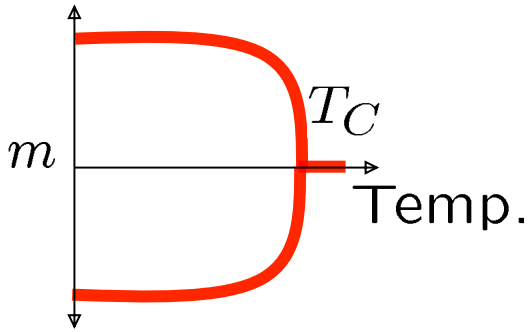
Storage

Long time behavior (not infinite time, but effectively)

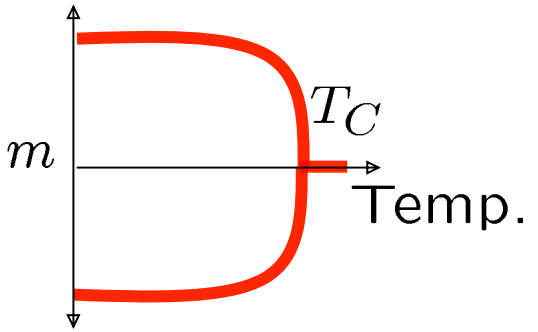
1D Ising



2D Ising



Toom's Rule



Criticism: (1) thermodynamic limit taken

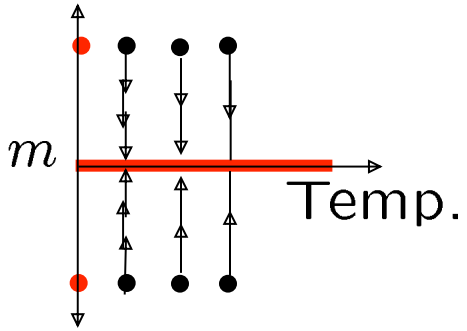
(2) what if relaxation to equilibrium takes a long time?

Storage



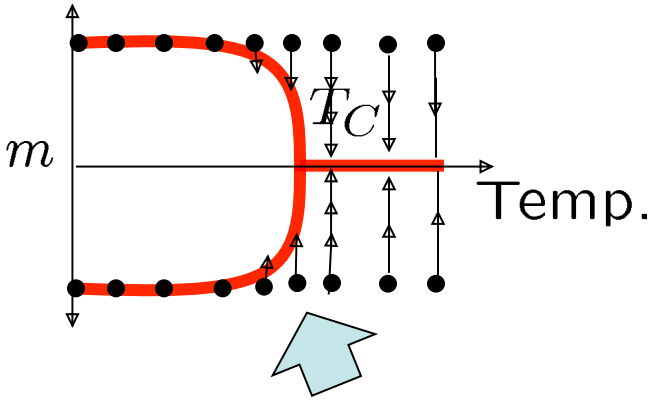
2nd attempt:
1D Ising

Relaxation to thermal



decay rate $\propto e^{-\beta J}$

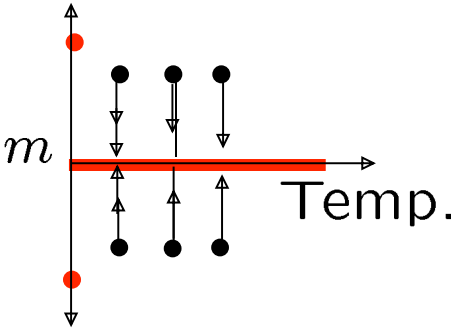
2D Ising, Toom's Rule

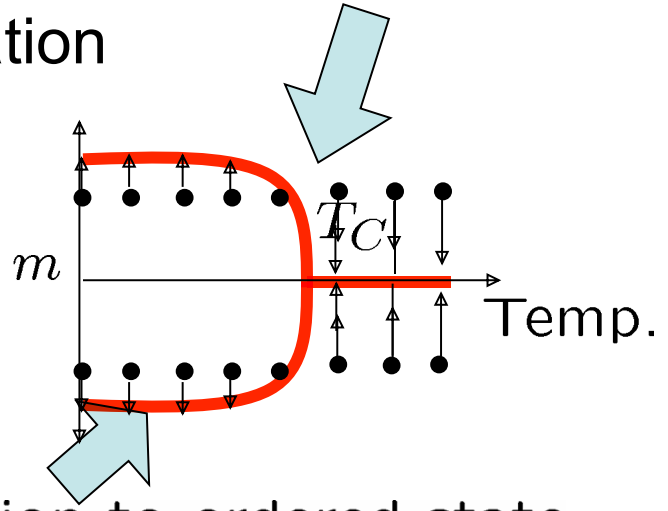


critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

Imperfect preparation





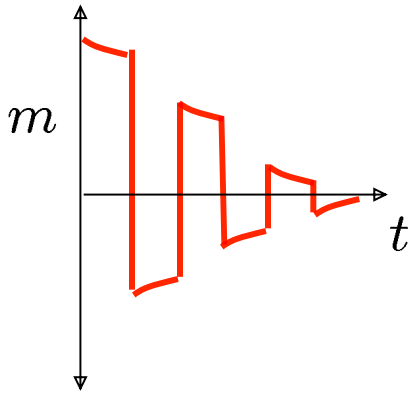
fast relaxation to ordered state

Manipulation



Flipping spins is imperfect

1D Ising

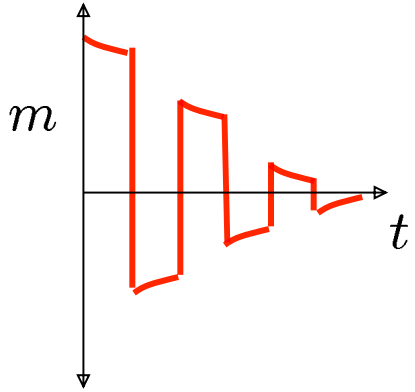


1D Ising
- temperature less than gap
- manipulation error must be small

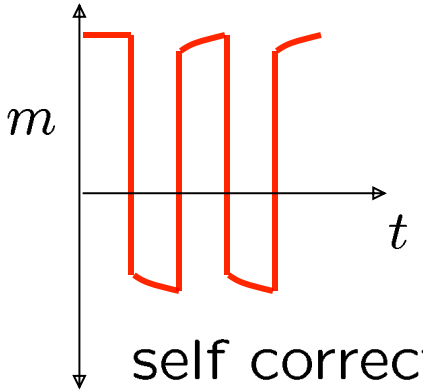
(i.e. NOT fault-tolerant)

2D Ising
- temperature less than critical temp

2D Ising
Toom's Rule



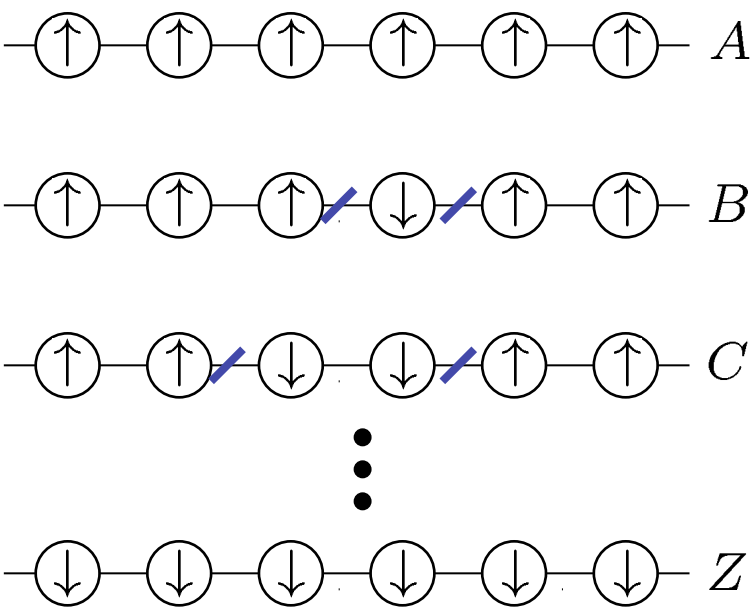
$T > T_C$

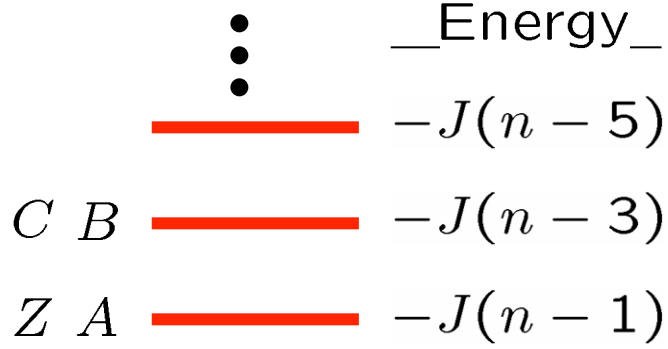


$T < T_C$

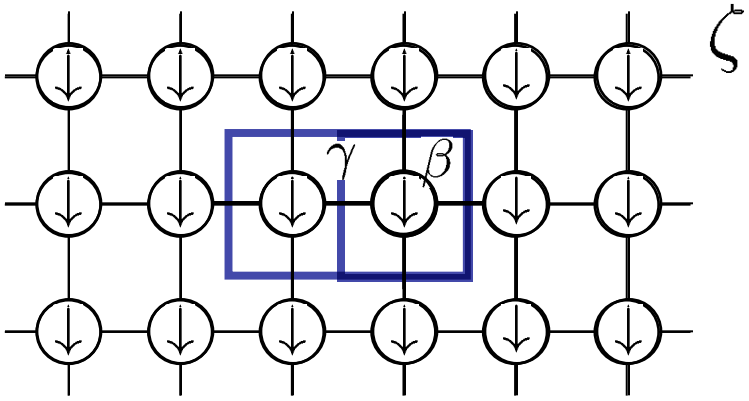
self correcting

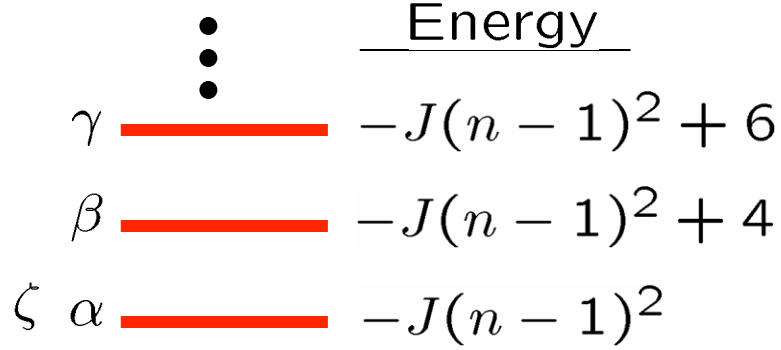
“Self Fixing”





Excitations point-like
Energy to err is constant
Disordering entropy >> Energy



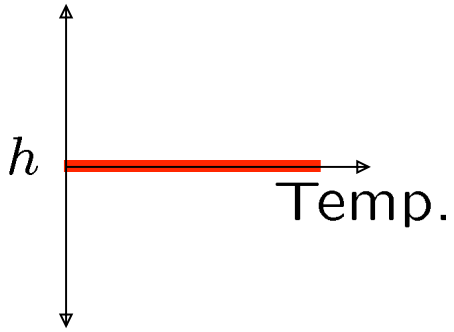


Excitations string-like
Energy to error is prop to size
Energy >> Disordering entropy

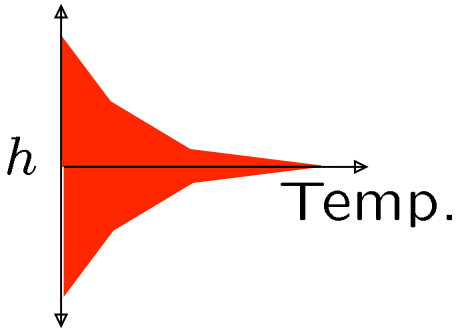
Toom's Rule Versus 2D Ising

Applied magnetic field (corresponds to a bias in noise)

2D Ising



Toom's Rule



Regions with two coexisting phases

External magnetic fields disorder magnetic media

But Still, a Technology?

Observation:



tum error correction is not that different from
ical error correction.

Observation:



Our most basic classical computational components
aturally robust (and often need only minor amounts
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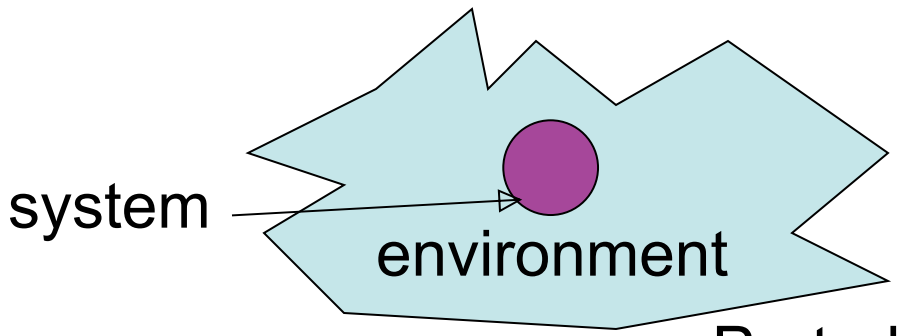
Observation:



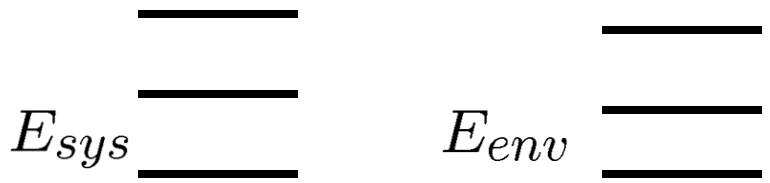
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Energetics of Decoherence

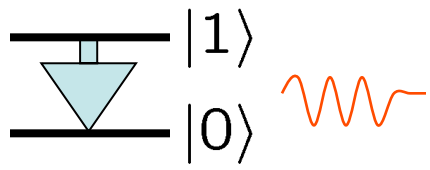


$$H = H_{sys} + H_{int} + H_{env}$$



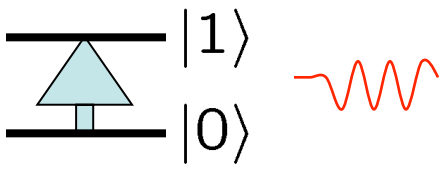
Perturbative regime: $E_{int} \ll E_{sys}, E_{env}$

Perturbative regime: decoherence conserves system and environment energy.



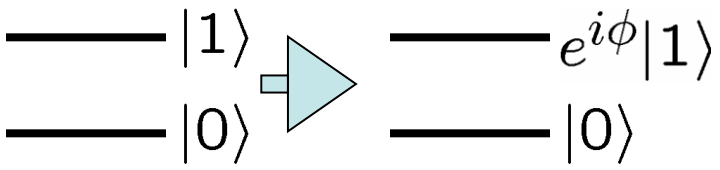
cooling

$\Delta E_{sys} < 0$



heating

$\Delta E_{sys} > 0$



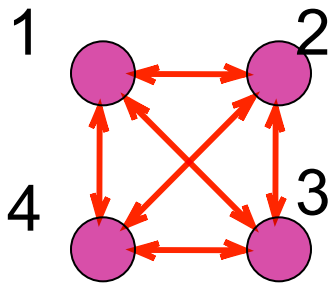
dephasing

$\Delta E_{sys} = 0$

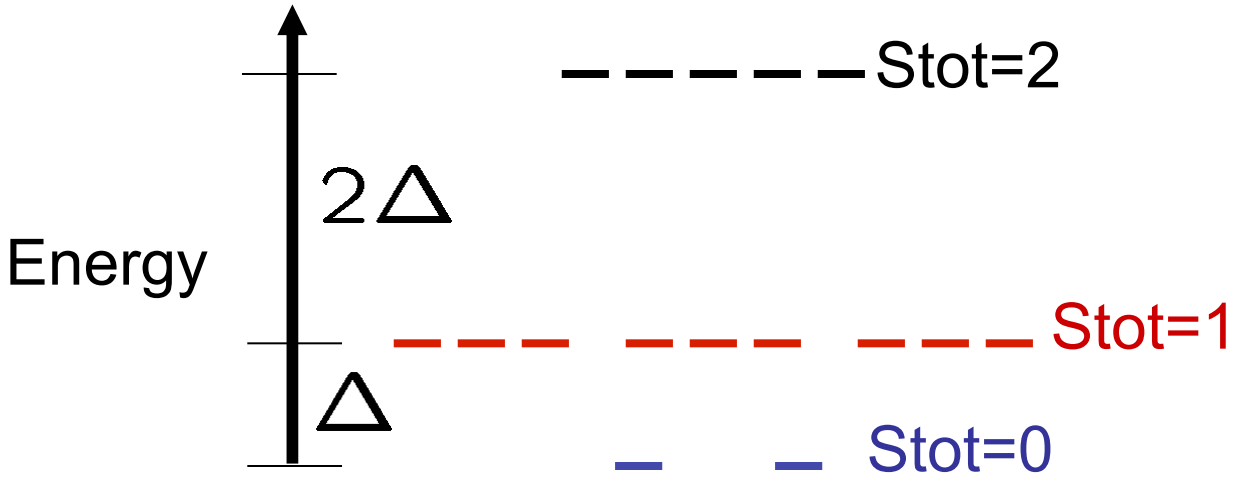
by **keeping a cool environment** we can eliminate heating

Exchange Everywhere

Can we engineer systems which turn dephasing/cooling → heating?



$$H = \Delta \sum_{i>j=1}^{4} \vec{S}_i \cdot \vec{S}_j = \frac{\Delta}{2}\left(S_{tot}^2 - 3I\right)$$



ground state is a quantum error correcting code

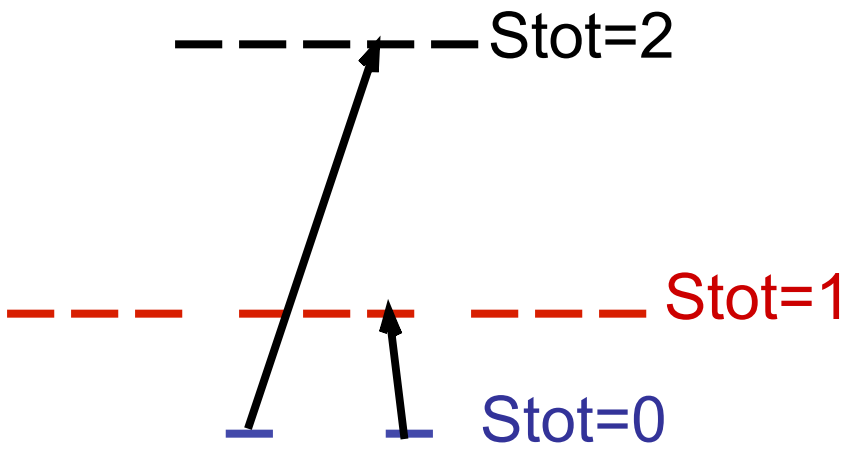
Encode quantum information into ground state

Supercoherence

[Bacon, Brown, and Whaley, *Phys. Rev. Lett.* 87, 247902 (2001)]

Independent coupling of system to environment

$$H_{int} = \sum_{i=1}^{4} S_\alpha^{(i)} \otimes B_\alpha^{(i)}$$



all single qubit interactions on this system are of heating (ground state is quantum error detecting code)

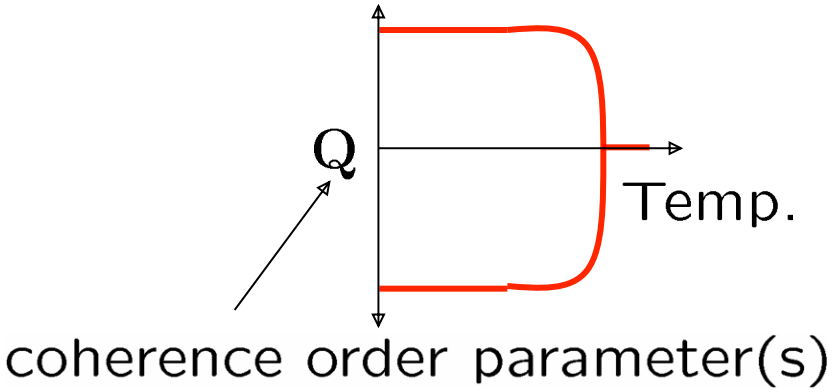
$S_\alpha^{(i)}$ never cause transitions on encoded space

All first order decoherence processes suppressed $kT \ll \Delta$

The Quantum Hard Drive?

RANT MODE ON

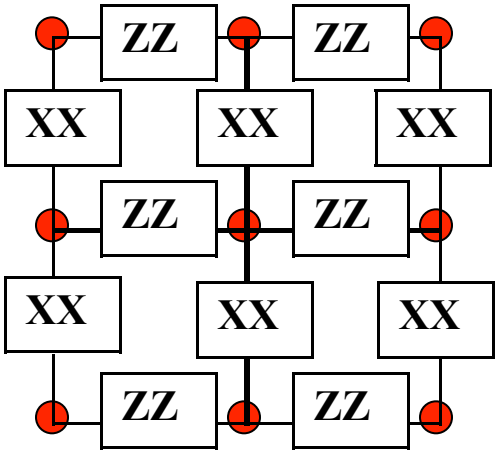
Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



Today in QC: we have theory (like von Neumann, Shannon) but not devices which implement these idea (without severe a device complexity overhead.)

Self-Correcting Quantum Computers

[Bacon, *Phys. Rev. A*, 73, 012340 (2006)]



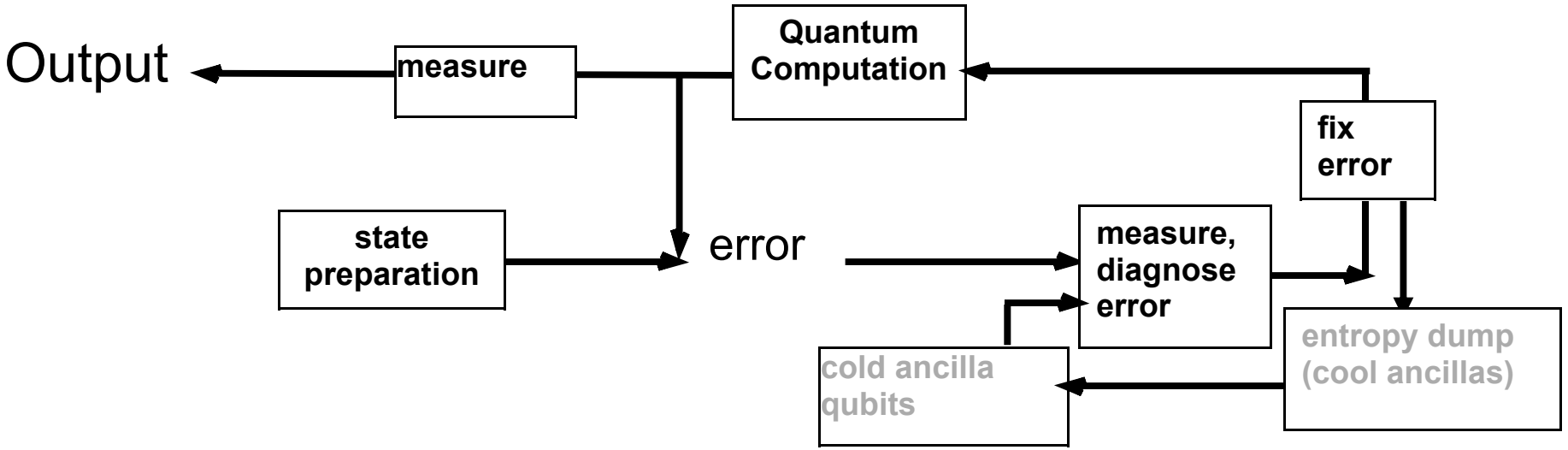
$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$

quantum compass model

Symmetries of this system kind of quantum error correcting:

quantum error correcting subsystem

Quantum Error Correcting Order Parameter

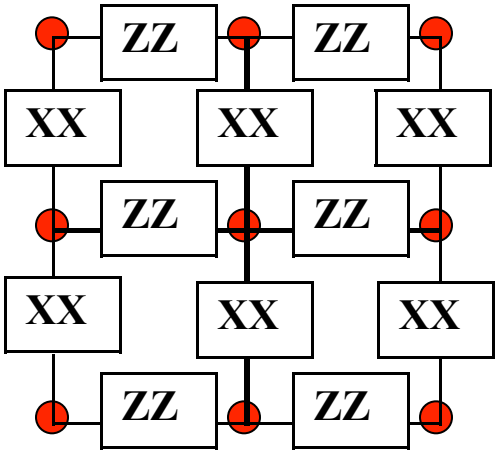


Erred quantum information in a quantum error correcting code is “still there.” How to see it? Perform correction, measure.

Quantum error correcting order parameter
(Actually central to Superconductivity, Quantum Hall effect,….)

Self-Correcting Quantum Computers

[Bacon, *Phys. Rev. A*, 73, 012340 (2006)]



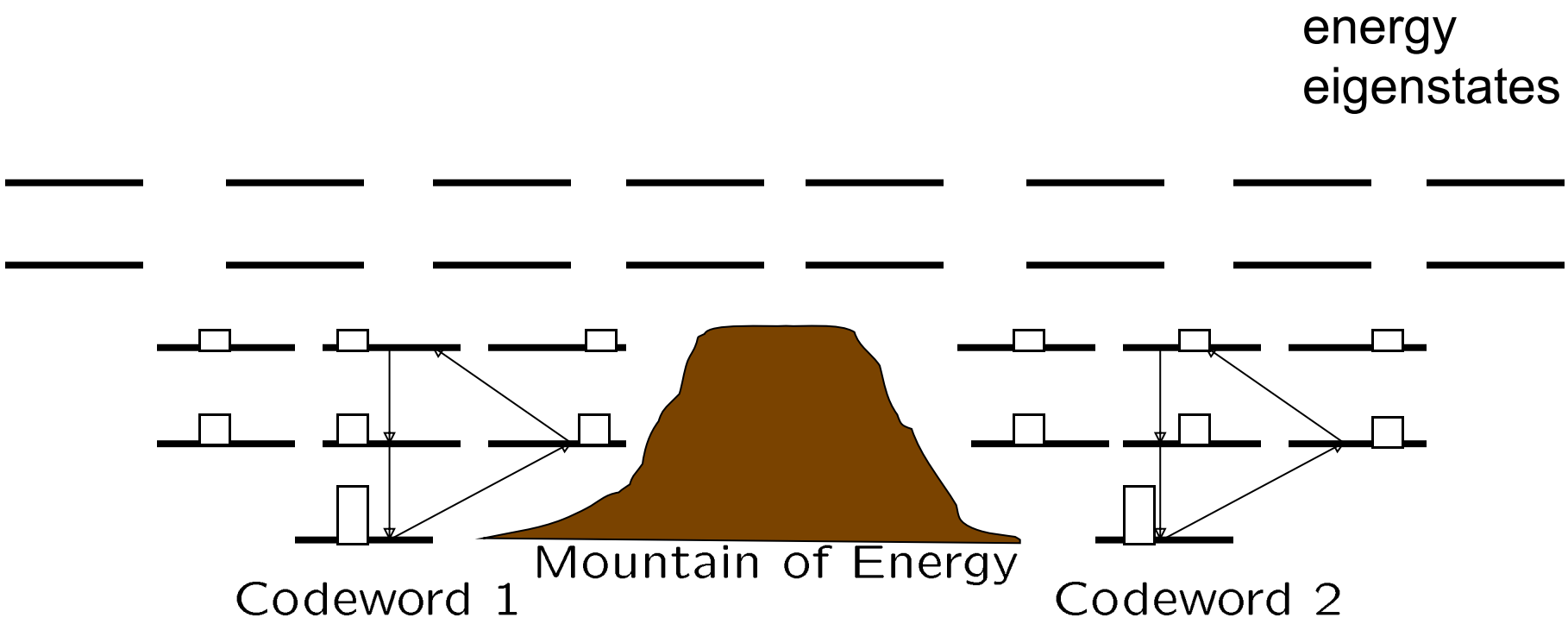
$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$

quantum compass model

Quantum error correcting procedure for quantum error correcting order parameter: surprise, subsystem code!

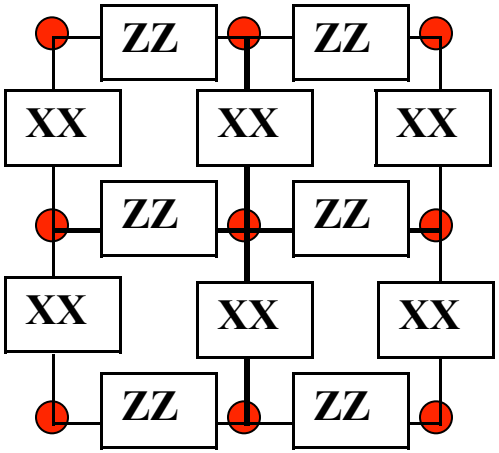
Spin-off: The code discovered here greatly reduces overhead for "traditional" fault-tolerant quantum computing (improves threshold) [Aliferas, Preskill, Cross 2006]

Self-Correction



Self-Correcting Quantum Computers

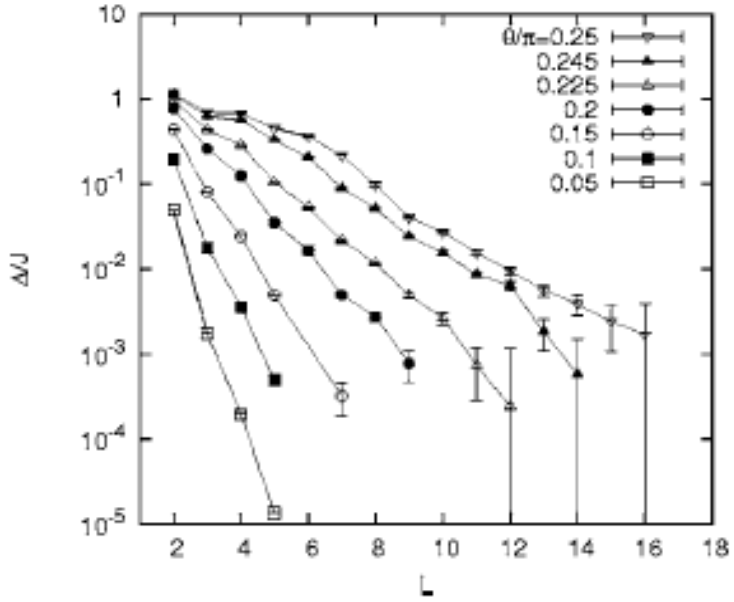
[Bacon, *Phys. Rev. A*, 73, 012340 (2006)]



$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$

quantum compass model

Self-correcting?



Energy gap vanishes for large lattice size Λ

Even worse, excitations are "particle like" Λ

[Dourier, Becca, Mila 05]

Ideas, Ideas, Ideas

[Bacon, *Phys. Rev. A*, 73, 012340 (2006)]
[Bacon, unpublished, 2006]

long range quantum compass model

$$H = -\sum_{i,j=1}^{n} \sum_{k=1}^{n-1} J_X(k) X_{i,j} X_{i+k,j} + J_Z(k) Z_{i,j} Z_{i,j+k}$$

doubled toric code combined + quantum compass model

$$\begin{aligned} H &= -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1} \\ &- J_t \sum_{ij} \left(X_{i,j} X_{i+1,j} X_{i,j+1} X_{i+1,j+1} + Z_{i,j} Z_{i+1,j} Z_{i,j+1} Z_{i+1,j+1} \right) \end{aligned}$$

three dimensional quantum compass model

$$H = -J_X \sum_{ijk} X_{i,j,k} X_{i+1,j,k} - J_Y \sum_{ijk} Y_{i,j,k} Y_{i,j+1,k} - J_Z \sum_{ijk} Z_{i,j,k} Z_{i,j,k+1}$$

Self-Correcting Quantum Computers

[Bacon, unpublished, 2006]

long range quantum compass model

$$H = -\sum_{i,j=1}^{n}\sum_{k=1}^{n-1} J_X(k) X_{i,j} X_{i+k,j} + J_Z(k) Z_{i,j} Z_{i,j+k}$$

First self correcting quantum memory!

Can we implement in ion traps?

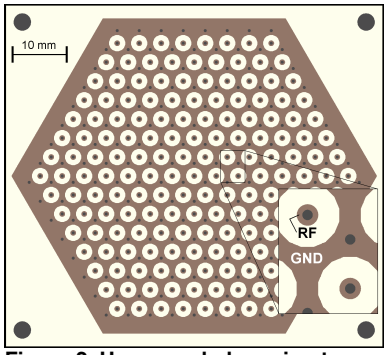
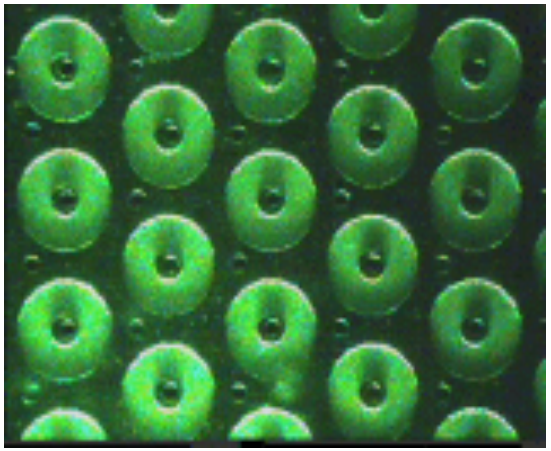


Figure 2. Hexagonal planar ion trap. 2D mult-zone ion traps allow one to trap ions in any spatial arrangement including a disordered lattice.



Pictures courtesy Ken Brown (Isaac Chuang's Lab, MIT)

Ken Brown ⇒ (Georgia Tech, joint Chemistry/CS)

Self-Correcting Quantum Computers

Strange meshing of:
- quantum error correction
- thermodynamics
- fault-tolerant quantum computation
- many-body quantum physics



NSF EMT Grant (with Mark Oskin)
“Self-Correcting Fault-Tolerant Quantum Computers”

Ph.D. students:
- Paul Pham (CSE), Gregory Crosswhite (Physics)

Masters students:
- Yoshiro Toda (Evening Masters, Physics)

Undergrad projects:
- Loren Merritt, Thomas Patecky

Where To?



Self-correcting quantum computers

- First proof of principle. Need more. How ubiquitous?
- Physical implementation (ion traps, optical lattices, superconducting circuits)
- Universal fault-tolerant constructions

End result: equivalent of the transistor/integrated circuit

Theory of quantum error correction

- New subsystem codes: subsystem codes bring quantum error correction even closer to classical error correction.
- General theory of fault-tolerance (current methods are ad hoc with few unifying principles.)

End results: aid in constructing self-correcting quantum computers and more traditional quantum error correcting microarchitectures



Why the Confidence?



Why are we confident that large scale quantum computers can be built?

Why are we confident that quantum computers are worth studying and building?



A historical account: including the future!

Stop the Myth

Quantum Algorithms

- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]
- Amplitude Amplification [Long List]
- Various hidden subgroup problems [Long List]
- Pell's equation [Hallgren 02]
- Hidden shift problems [van Dam, Hallgren, Ip 03, Childs, van Dam 05]
- Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]
- …

What Worth, Quantum Computation?

Quantum algorithms research is about much more than just coming up with new quantum algorithms.

Bold Reasons

Suppose that tomorrow someone proves that P≠NP, would we be satisfied? BQP≠NP?

The language of the universe is quantum mechanical, shouldn't the language of our theory of computation match?

[Kerenidis, de Wolf 02] Length bounds on 2 query locally decodable codes via a quantum proof

[Aaronson 04] PP closed under intersection via post selected quantum computation

[Kerenidis 05] Classical circuit lower bounds from quantum communication complexity bounds

What Worth, Quantum Computation?

Quantum algorithms research is about much more than just coming up with new quantum algorithms.

Bold Reasons

Understanding what “powers” quantum algorithms helps us come up with new algorithms, but it also helps us understand when a quantum system can be simulated

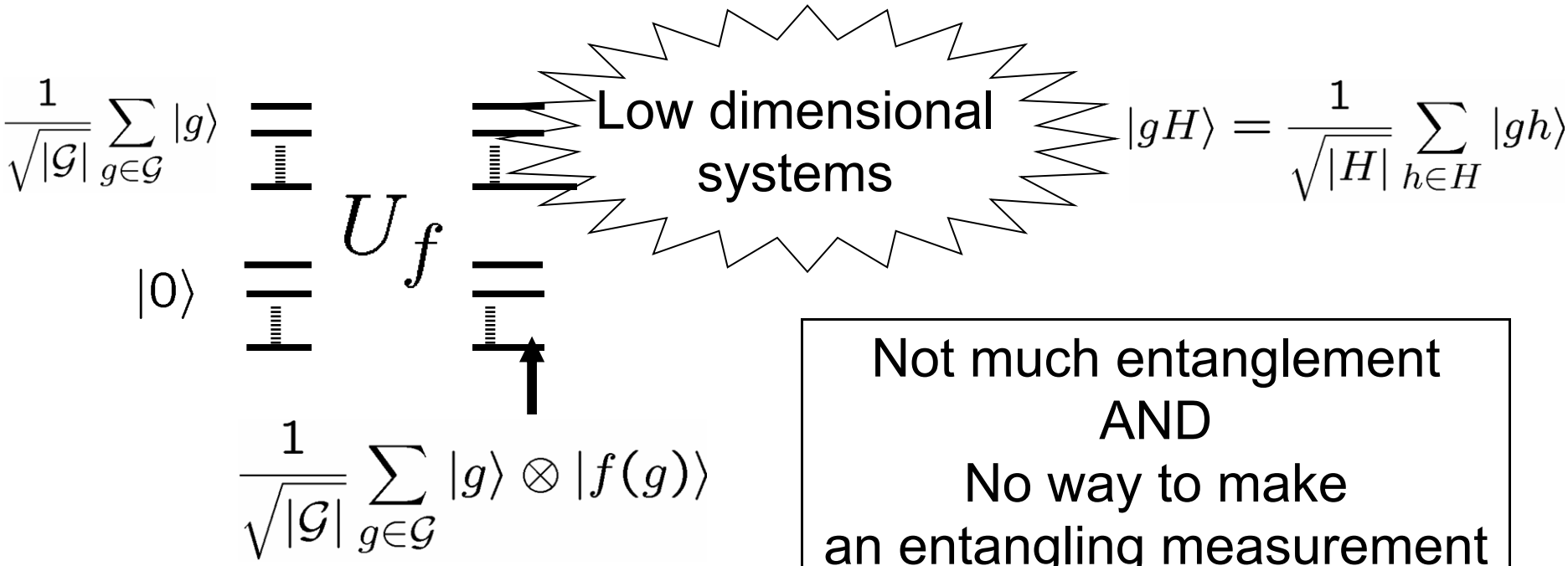
Entanglement is necessary for quantum speedups

[Jozsa and Linden 02]

New (classical) algorithms for efficient simulation of certain many-body quantum systems

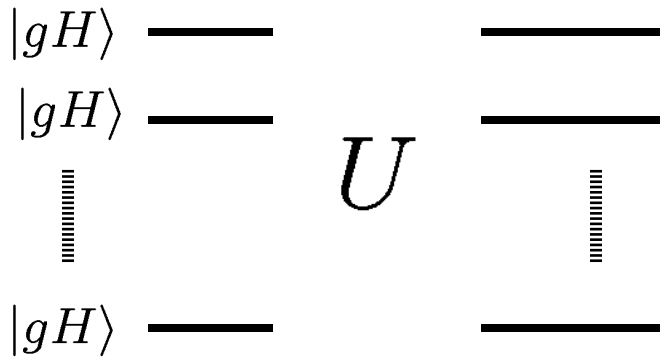
[Vidal 03, Verstraete, Cirac 05]

Example: Hidden Subgroup Problem



Not much entanglement
AND
No way to make
an entangling measurement

k copies of the hidden subgroup states:



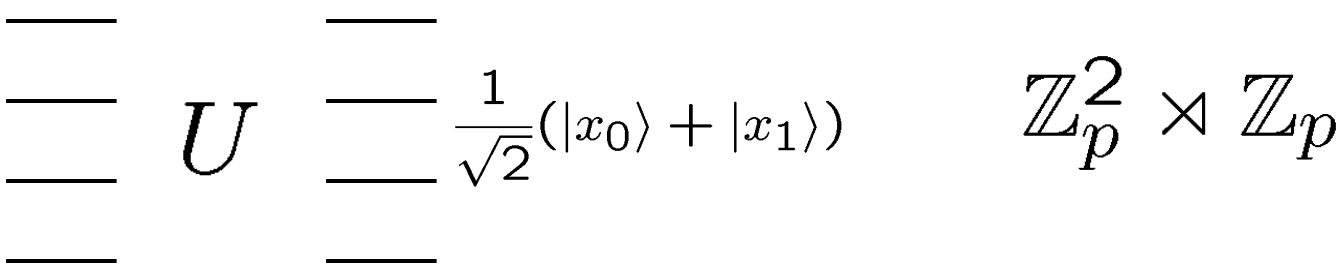
entangled
measurement
to determine the
hidden subgroup

Example: Hidden Subgroup Problem

Bacon, Childs, van Dam FOCS 2005

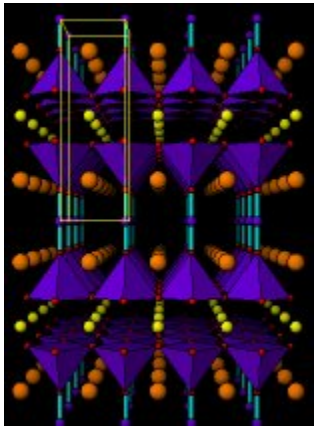
"From Optimal Measurement to Efficient Quantum Algorithms for the Hidden Subgroup Problem over Semidirect Product Groups"

Basic new powerful primitive? Ability to produce superposition of solutions to quadratic equations (inverse of this transform.)



What Worth, Quantum Computation?

Quantum algorithms research is about much more than just coming up with new quantum algorithms.



Bold Reasons

Simulating quantum systems
(biomolecular, quantum solids)

Quantum materials
Simulation, verification, design, understanding….

Real time feedback in quantum materials: currency of quantum information in design of quantum materials with novel physical behavior

The Coming Quantum Age

• Every day experiments bring us closer and closer to viable large scale quantum computers
•Optimism is based on a beautiful robust theory of quantum error correction
• Optimism is based on comparisons of quantum theory with stochastic classical theory
• Ideas like **self-correcting quantum computers** will lead to quantum computers sooner rather than latter

• Quantum algorithms are the tip of a huge iceberg of ideas with tremendous potential for theory
• Quantum algorithms are vital to the future engineering of "quantum" materials
• **Entanglement** can guide in the construction of quantum algorithms

Towards Robust and Powerful Quantum Computers

Dave Bacon
University of Washington
Department of Computer Science & Engineering

